

word2vec

$$P(t | c) = \frac{e^{\theta_t^T e_c}}{\sum_{j=1}^N e^{\theta_j^T e_c}}$$

target content content embedding target embedding

N is very large $\rightarrow$ Hierarchical softmax
 $\rightarrow$ Negative sampling

$$\max_{e, \theta} \left[\log \sigma(\theta_t^T e_c) + \sum_{j \neq t} \log \sigma(-\theta_j^T e_c) \right]$$

sampling by raising unigram freq
to $3/4$ power